

Project Presentation of Industry Oriented Hands on Experience (22CS421)

On

Crop Yield Prediction Using Machine Learning

Aditya Mehta

2210990068

Supervised By

Dr. Gurpreet Singh

Department of Computer Science and Engineering,
Chitkara University, Punjab



Introduction: Why Crop Yield Prediction Matters

The Challenge

- **Food security:** India feeds 1.4 billion people; unpredictable yields threaten supply chains
- **Climate variability:** Erratic monsoons, temperature extremes, and shifting seasons disrupt traditional forecasting
- **Precision agriculture:** Data-driven decisions can reduce waste and optimise inputs

Why Machine Learning?

Non-linear Patterns

Captures complex agro-climatic interactions invisible to statistical models

Multi-feature Fusion

Integrates climate, soil, and management data simultaneously

Scalability

Generalises across crops, states, and seasons without re-parameterisation

Problem Statement



Traditional Methods

- Statistical Regression
- Expert Heuristics
- Limited Feature Sets
- Single-Crop Focus
- Poor Generalisation



Machine Learning Approaches

- Ensemble Models
- Data-Driven Patterns
- Multi-Feature Integration
- Multi-Crop Benchmarking
- High Accuracy across Regions

Key Gaps Identified

- Traditional statistical models fail to capture **non-linear agro-climatic relationships**
- No standardised **Indian agro-climatic benchmark dataset** exists for multi-crop comparison
- Prior studies lack **systematic benchmarking** across six or more ML algorithms
- Single-crop studies cannot generalise to India's diverse agro-ecological zones

⚠ A reproducible, multi-crop ML benchmark for Indian agriculture is critically absent from current literature.



Research Objectives

1

Compare 6 ML Algorithms

Benchmark Ridge Regression, Decision Tree, Random Forest, SVR, XGBoost, and Gradient Boosting under identical conditions

2

Predict Crop Yield Across Indian States

Cover 15 states, 6 major crops, and 26 years of historical agro-climatic data

3

Evaluate Model Accuracy

Use MAE, RMSE, R^2 , MAPE, and CV- R^2 for rigorous, multi-metric evaluation

4

Analyse Feature Importance

Identify the most predictive agro-climatic, soil, and management features

5

Build Reproducible Benchmark

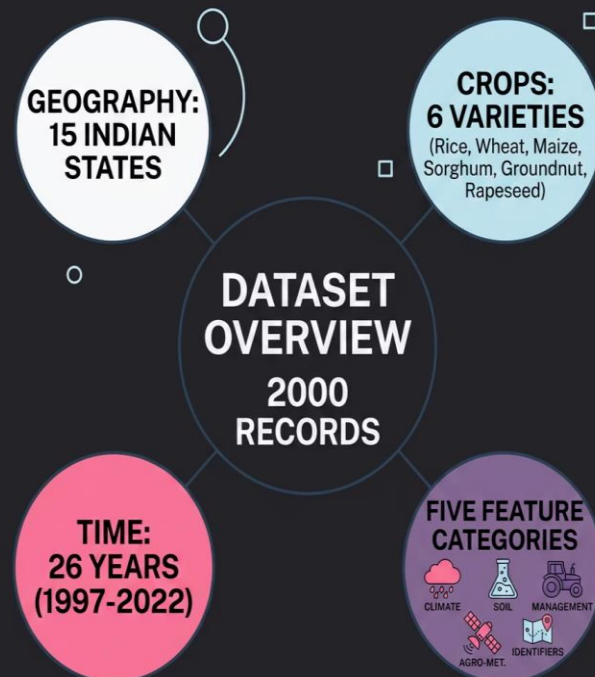
Release a standardised pipeline for future Indian agricultural ML research

Literature Review

Existing studies demonstrate the promise of ML for yield prediction but reveal critical gaps — particularly in multi-crop, multi-region Indian benchmarking.

Study	Algorithm	Region / Crop	Limitation	Gap Addressed Here
Pantazi et al.	SOM, MLP	Greece / Wheat	Single crop, small dataset	Multi-crop benchmarking
Jeong et al.	Random Forest	USA / Maize	No soil features	26 integrated features
Khaki & Wang	Deep Learning	USA / Soybean	High compute, low interpretability	Lightweight ensemble models
Schwalbert et al.	ML ensemble	Brazil / Soybean	Single country, single crop	15 Indian states, 6 crops
Gandhi et al.	Regression	India / Mixed	No standardised dataset	Reproducible benchmark pipeline

Dataset & Features



Feature Categories



Climate

Rainfall, temperature, humidity, sunshine hours



Soil

pH, N, P, K, organic carbon content



Agro-met Indices

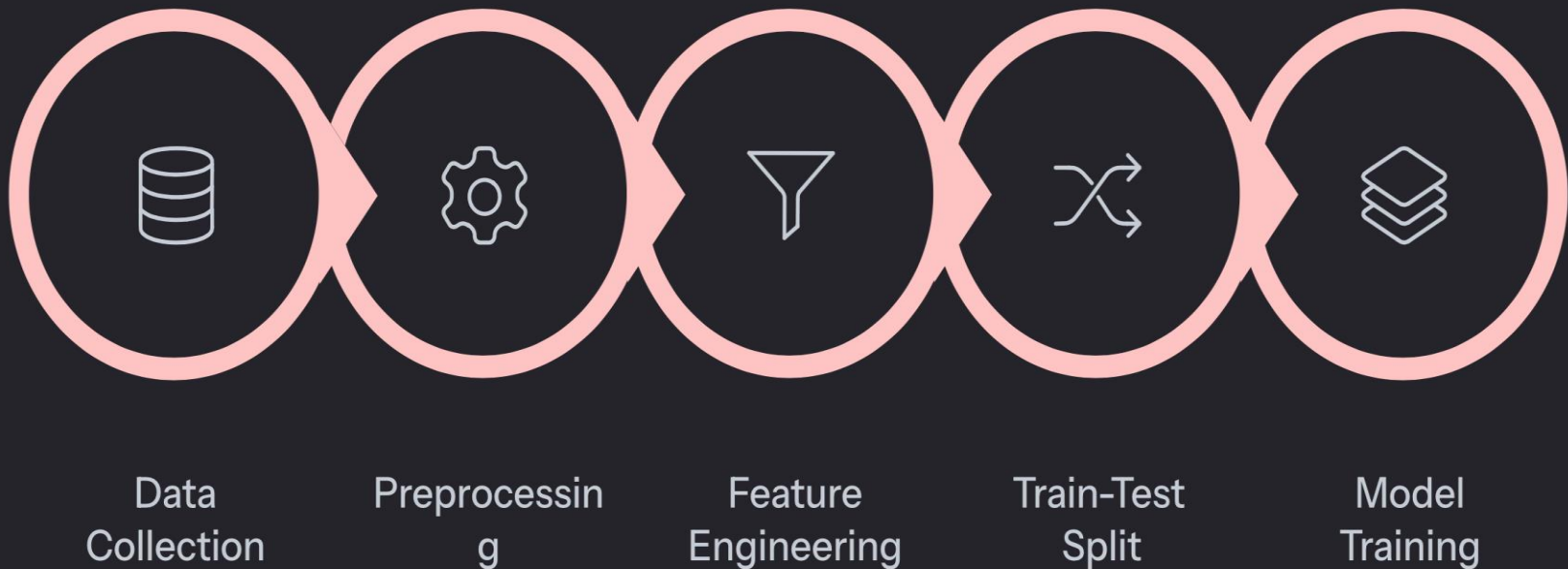
Peak NDVI, ETo, growing degree days



Management

Irrigation depth, fertiliser application rate

Methodology: ML Pipeline



A temporal train-test split (80/20) was enforced to prevent data leakage. StandardScaler normalised continuous features, while Label Encoding handled categorical variables. Five-fold cross-validation ensured robust generalisation across unseen agro-climatic conditions.

Machine Learning Models



Ridge Regression

Linear baseline with L2 regularisation. Simple, interpretable, but limited to linear relationships.



Decision Tree

Rule-based splits on feature thresholds. Interpretable but prone to overfitting on noisy data.



Random Forest

Ensemble of decorrelated trees. Robust to noise; strong generalisation via bagging.



SVR

Margin-based regression using RBF kernel. Effective in high-dimensional spaces; computationally intensive.



XGBoost

Regularised gradient boosting. High accuracy with built-in feature importance and sparsity handling.



Gradient Boosting

Sequential residual correction. Best-performing model; excels at capturing complex non-linear patterns.



Results: Model Performance Comparison



Key Results

🏆 Gradient Boosting

$R^2 = 0.9247$ · $RMSE = 0.2664$ · $MAPE = 4.31\%$

XGBoost

$R^2 = 0.9228$ — close second, strong regularisation

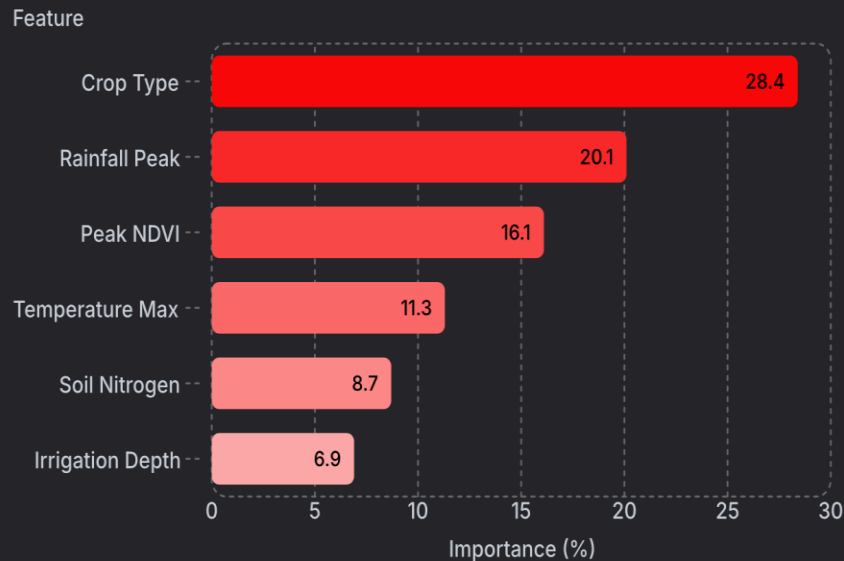
Ridge Regression

$R^2 = 0.4341$ — confirms linearity is insufficient for agro-climatic data

✓ Ensemble methods outperform linear and single-tree models by a significant margin, validating the non-linear nature of crop yield dynamics.

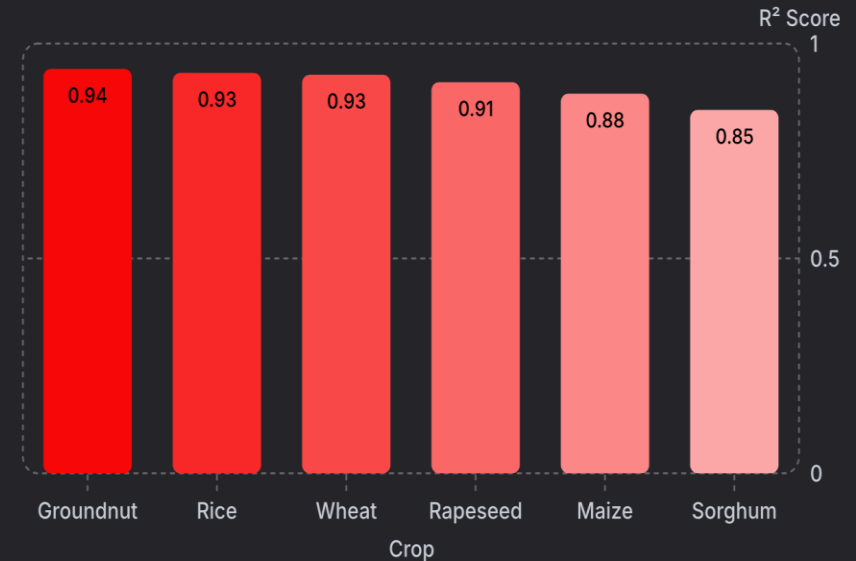
Feature Importance & Crop-wise Performance

Top Predictive Features



📌 Crop Type, Rainfall Peak, and Peak NDVI together account for **64.6%** of total predictive variance.

Crop-wise R^2 Performance



Groundnut achieved the highest R^2 (0.9412); **Sorghum** showed the lowest, likely due to greater agro-climatic variability across states.



Thank You